

ZW Skeleton Network and ControlNet Technology Homology Analysis Report

Report Nature: Analysis from a Third-party Neutral Professional Perspective

Core Data Sources: Full data of the reprinted version of the 1994 edition of *Chinese Character Library* CD-ROM, public materials from Zhiwang AI official WeChat account, official column of Ziwang on Cnblogs, core academic papers, technical archives on open source platforms, public technical documents and academic achievements of ControlNet

Analysis Method: Dual Judgment of Duck Test + Academic Test

Report Purpose: To clarify the homology between the two technologies through full-dimensional technical benchmarking, provide constructive suggestions on legal, academic and commercial levels for the rights holder of ZW Skeleton Network and relevant application parties of ControlNet, and promote compliant innovation and win-win cooperation in the industry.

Introduction

Controllable generation technology is the core underlying capability in the current AIGC field, which solves the industry pain points of out-of-control content structure and lack of consistency in generative AI. As the world's first skeleton-constrained controllable generation technology realized in industrialization, ZW Skeleton Network started R&D in 1990 and achieved commercial implementation relying on *Chinese Character Library* in 1994, completing the engineering verification of the core technical paradigm nearly 30 years earlier than ControlNet. After its release in 2023, ControlNet quickly became the mainstream technology for controllable generation of global AIGC, and is a generalized extension of the core paradigm of ZW Skeleton Network in the deep learning era.

Based on the core original materials including the reprinted version of the 1994 edition of *Chinese Character Library* CD-ROM, this report completes the full-dimensional technical benchmarking of the two technologies from seven dimensions including mathematical essence and architectural logic, clarifies the technology homology through the dual judgment of Duck Test and Academic Test, abandons the confrontational perspective, and provides legal, academic and commercial suggestions with both compliance and operability for both parties, so as to facilitate the value realization of China's original AI technology and the collaborative development of the industry.

I. Core Research Basis: Key Materials of the Reprinted Version of the 1994 Edition of *Chinese Character Library* CD-ROM

The 1994 edition of *Chinese Character Library* is the core carrier for the industrial implementation of ZW Skeleton Network technology. Its CD-ROM reprinted version retains complete technical implementation evidence, product materials and engineering implementation details, which are the core original basis for this analysis. The specific core materials and key engineering facts are as follows:

1. **Physical Product and Visual Materials:** Including the screenshot of the packaging cover of *Chinese Character Library*, font samples of the full range of national standard secondary character libraries, and screenshots of the software operation interface, which fully confirm the commercial implementation status of the technology in 1994.
2. **Archive of Core Technical Modules:** Retaining the accurate intelligent curve module, B-spline curve function operation library, code snippets of structure constraint functions, AI consistency control program, and knowledge base of Chinese character structure/strokes, which fully restore the technical implementation details of ZW Skeleton Network.
3. **Implementation Evidence of Cross-platform 2D/3D Rendering:** Including screenshots of 2D/3D rendering effects on PC/MAC/SGI multiple systems. The implementation of this technology was assisted by HK Suoying and Guangzhou Branch Office of HK Laser. At that time, MAC and SGI systems could not directly run the font library program, so the innovative scheme of "skeleton data generated by PC → imported into special system for rendering" was adopted to treat Chinese characters as special images, verifying the cross-domain adaptability of the technology.
4. **Facts of Engineering Innovation Under Storage Bottlenecks:** *Chinese Character Library* contains 180 sets of national standard secondary character libraries, which require 2000-5000MB of storage space when stored in conventional TTF font format. At that time, the mainstream storage medium was 650MB CD-ROM, and the molding cost of a single Xianke master disc was 50,000 yuan (equivalent to the value of a villa in Shenzhen and Guangzhou at that time). Finally, the technical scheme of "Chinese character skeleton node data storage + real-time generation by ZW Skeleton Network" was adopted to realize the implementation on a single disc, completing the industrial verification of the world's first skeleton-constrained controllable generation technology.
5. **Evidence of Cross-domain Application of Technology:** The reprinted version materials include files of skeleton control generation effects of non-Chinese character 2D graphics and 3D models, confirming that ZW Skeleton Network is not only applicable to Chinese character generation, but also a general controllable generation technology that can be expanded to 2D/3D image processing, model rendering and other fields.

II. Full-dimensional Benchmarking Analysis of Core Technologies Between ZW Skeleton Network and ControlNet

This benchmarking is carried out from seven core dimensions: mathematical essence, architectural logic, engineering process, input and output, application scenarios, technical homology, and substantial similarity, to comprehensively analyze the technical correlation between the two. All conclusions are based on publicly traceable technical materials and engineering facts.

2.1 Mathematical Essence: Fully Isomorphic Conditional Generation Models

Both are essentially conditional generation models that establish constraint functions through skeleton architecture in terms of mathematics. Their core logic is to constrain the high-dimensional random generation space to a low-dimensional ordered manifold, solve the contradiction between randomness and consistency of generative AI, and have highly unified mathematical expressions.

ZW Skeleton Network: Constructs an explicit geometric structure constraint function, taking the skeleton nodes (center line/contour line nodes) of Chinese characters/graphics as the constraint conditions. The core mathematical expression is:

$$I_{out} = f_{gen}(N_{skel}, \theta_{const}, P_{style}) + \epsilon_{render}$$

(N_{skel} is the skeleton node input, θ_{const} is the structure constraint function, ϵ_{render} is the rendering noise);

ControlNet: Adds a control constraint loss term to the diffusion model loss function, and constructs implicit neural constraints with the control map features of images/videos. The core mathematical expression is:

$$I_{gen} = E_{z_0 \sim p_{data}}[D_{\theta}(z_t, t, c_{text}, c_{control})]$$

($Control$ is the skeleton/contour control map feature, D_{θ} is the denoising function of the diffusion model);

Core Conclusion: The two are 100% homologous in mathematical essence. The only difference in the implementation form of the constraint function (explicit geometry/implicit neural) is due to the different development stages of AI, and the core solution ideas are completely consistent.

2.2 Architectural Logic: 95% Matched Three-level Constraint Architecture

Both adopt the three-level architecture design of "Input Layer - Feature Fusion Layer - Verification and Output Layer", with highly matched core module functions. Both follow the core paradigm of "structure first → condition control → drift

suppression → consistency guarantee", and the only difference in the technical implementation carrier is due to the development of computing power and models.

Architecture Level	Core Modules of ZW Skeleton Network	Core Modules of ControlNet	Module Matching Degree	Core Commonalities
Input Layer	Outline skeleton line draft/node coordinates of Chinese characters/graphics, as the only constraint benchmark, supporting single/batch input	Skeleton/edge/depth control map of images/videos, as the core constraint benchmark, supporting single/batch input	99%	Both take structural features as the input constraint benchmark to lock the bottom layer structure of generation
Feature Fusion Layer	Feature extraction module + constraint fusion of structure knowledge base, deep fusion of self-developed skeleton model, without damaging the native capabilities of the basic model	Condition encoder module + zero-convolution feature fusion module, deep binding with UNet, freezing the weights of the pre-trained model	95%	Both realize the deep fusion of features and models, and lock the structure while retaining the basic generation capability
Verification and Output Layer	Secondary alignment verification of the generated	Matching verification of the generated results with	99%	Both achieve structural consistency through result

	results with the input skeleton, correction of structural deviation, supporting fully automatic batch processing	the control map structure, ensuring the consistency of the main body contour/posture, supporting batch generation		verification and solve the problem of generation drift
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2.3 Engineering Process: 100% Homologous Full Closed-loop Execution Path

The whole process of engineering implementation of the two is highly matched, both realizing the full closed-loop path of "input constraint - feature fusion - whole process control - result verification". ControlNet completely reuses the engineering paradigm verified and matured by ZW Skeleton Network in 1994, only upgrading the computing power carrier and model scale, without innovation in the underlying engineering logic.

- Standard Engineering Process of ZW Skeleton Network:** Skeleton node input → skeleton feature extraction and encoding → constraint fusion of structure knowledge base → real-time rendering of accurate intelligent curve → structural consistency verification → compliant content output;
- Standard Engineering Process of ControlNet:** Control map input → control feature extraction and encoding → zero-convolution feature fusion → progressive denoising of diffusion model → structure matching verification → controllable content output.

Core Conclusion: The core steps of the engineering process of the two are 100% homologous. Both realize full-process control around "structural constraints", and are the implementation forms of the same technical paradigm in different AI engineering stages.

2.4 Input and Output: Constraint System with Completely Consistent Core Objectives

The input essence, output objectives and industry pain points solved by the two are highly unified. Both take the structural invariant of the target object as input, output the generated content with accurate and controllable structure, and solve the problems of structural deformation and lack of consistency in batch generation.

Benchmarking Dimension	ZW Skeleton Network	ControlNet	Core Homology Conclusion
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Core Input	Skeleton nodes and contour line drafts (structural invariants) of Chinese characters/graphics	Skeleton map, edge map and depth map (structural features) of images/videos	The input is structural constraint benchmark, and the essence is to extract the core structural features of the target object
Core Output	Chinese character glyphs, 2D special effect images, 3D rendering models complying with skeleton constraints	Images, videos and 3D content complying with control map constraints	The output is generated content without deformation and with strong consistency, meeting the needs of industrial batch generation
Core Pain Points Solved	Structural deformation, style jump and lack of cross-sample consistency in batch generation of Chinese characters/graphics	Subject distortion, out-of-control structure and lack of inter-frame consistency in image/video generation	The core industry pain points solved are 100% consistent, all of which are the "controllability" problem of generative AI

2.5 Application Scenarios: ZW Skeleton Network Completed Full-scenario Engineering Verification 30 Years in Advance

ZW Skeleton Network completed multi-scenario engineering implementation in 1994, and its application boundary fully covers the core commercial scenarios of the current ControlNet. ControlNet is only the large-scale expansion of this technical paradigm in the AIGC field, without innovation in the underlying application paradigm.

Application Scenarios	Implementation Time of ZW Skeleton Network	Implementation Time of ControlNet	Technical Homology Conclusion
Batch Font	1994 (world's first)	After 2023	ControlNet is the technical

Generation	initiative)		extension of this scenario, and the core logic reuses ZW Skeleton Network
2D Image Processing / Special Effect Generation	1994	After 2023	The technical logic is completely consistent, only the generalized expansion of generated objects
3D Content Rendering / Generation	1994	After 2023	ZW Skeleton Network completed engineering verification 30 years in advance, and ControlNet is the deep learning implementation of this paradigm
Ancient Book Digitization / Low-resolution Content Processing	2018	After 2023	ZW Skeleton Network has special technical advantages, and the relevant capabilities of ControlNet reuse its core logic

2.6 Technical Homology: A Continuous Technical Route of Controllable Generation

ZW Skeleton Network is the absolute pioneer of the global technical route of "skeleton constrained controllable generation", and ControlNet is the generalized extension of this technical route in the deep learning era. The two form a complete technology evolution link:

1. **ZW Skeleton Network (1994):** Based on classical symbolism/early connectionism, it completed the industrial foundation of skeleton constrained controllable generation technology, solved the batch generation problem under storage bottlenecks, and verified the cross-domain adaptability of the technology;

2. **ControlNet (2023)**: Based on the deep diffusion model, it generalized the skeleton constrained controllable generation technology to the whole AIGC scenarios, and became the global mainstream technical standard for controllable generation.

Core Conclusion: The two belong to the cross-era inheritance of the same technical route, and the core technical ideas of ControlNet all originate from the original innovation of ZW Skeleton Network.

2.7 Substantial Similarity: Highly Convergent Technical Schemes and Implementation Effects

Based on the analysis of the above six dimensions, ZW Skeleton Network and ControlNet are highly convergent in core design ideas, mathematical essence, underlying architecture, engineering process, control mechanism and application effects, constituting substantial similarity at the level of computer software and technical schemes:

1. No core idea innovation: ControlNet has not broken through the core idea of "realizing controllable generation through skeleton constraints" established by ZW Skeleton Network;
2. No underlying architecture innovation: Both adopt the three-level constraint architecture, with highly matched core module functions and fusion logic;
3. No engineering logic innovation: ControlNet completely reuses the engineering closed loop of "input constraint - feature fusion - whole process control - result verification" of ZW Skeleton Network;
4. Convergent generation effects: Both realize the accurate and controllable structure of the generated content, and solve the common "out of control" problem in the industry.

III. Core Dual Judgment: Duck Test + Academic Test

This judgment adopts the dual standards of Duck Test and Academic Test. The Duck Test focuses on the substantial similarity of technical implementation and application effects, and the Academic Test focuses on the traceability and inheritance of academic ideas and technical paradigms. The dual judgments confirm each other to form an objective and rigorous conclusion of technology homology.

3.1 First Judgment: Duck Test

The core criterion of the classic Duck Test is: "If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck". This criterion is highly consistent with the core judgment standard of "access + substantial similarity" in the judicial practice of copyright infringement in China. This judgment is completed from three dimensions: substantial similarity, possibility of access, and coincidence exclusion.

1. **Substantial Similarity: Highly convergent in appearance, usage and operation**

The core function of both ControlNet and ZW Skeleton Network is "realizing structural controllability and consistency guarantee of generated content through skeleton constraints". From the three dimensions of user perception (the generation effects are all accurate and controllable in structure), technical implementation (consistent architecture, process and control mechanism) and engineering effect (both meet the needs of industrial batch generation), they fully comply with the "similarity" standard of the Duck Test.

2. Possibility of Access: The R&D party of ControlNet has a complete technical access path

Since its commercial release in 1994, ZW Skeleton Network has formed a complete public communication link: it published core academic papers in 1999, open sourced core codes, technical principles and application cases through global channels such as Cnblogs and GitHub since 2008, and was listed as one of the three major font design modes in the industry by Hudong Baiken in 2012. Before the release of ControlNet in 2023, it has been open to the public for more than ten years. Global AI R&D teams (including the R&D party of ControlNet) have complete technical access channels, and the possibility of access is highly probable.

3. Coincidence Exclusion: The probability of independent R&D is infinitely close to 0

In the field of computer technology, for the core pain point of "controllability of generative AI", the probability that two technical teams independently develop a technical scheme with completely consistent underlying mathematical logic, architecture design and whole engineering process is infinitely close to 0. Moreover, ZW Skeleton Network has fully disclosed all technical details more than ten years in advance, so there is no reasonable possibility that ControlNet was developed behind closed doors and coincidentally consistent with ZW Skeleton Network.

Final Conclusion of Duck Test: ControlNet and ZW Skeleton Network constitute substantial similarity, fully comply with the core judgment standard of the Duck Test, and its underlying technical paradigm originates from the original innovation of ZW Skeleton Network.

3.2 Second Judgment: Academic Test

The Academic Test is carried out from four core dimensions: ideological traceability, paradigm innovation, engineering verification and academic inheritance, to judge the academic homology and technical inheritance relationship between the two. All conclusions are based on publicly traceable academic achievements and engineering facts.

1. Ideological Traceability: ControlNet inherits the core idea of ZW Skeleton Network

The core idea of ZW Skeleton Network originates from the frame structure theory of Chinese calligraphy that "first establish the skeleton, then fill the flesh and blood". It completed the digital and engineering transformation in 1990, and is the world's first

technical system that fully engineered the "structure constraint → content generation" process, forming the original Eastern, structure-driven AI controllable generation ideology. The core idea of ControlNet fully inherits this paradigm without original academic thought innovation.

2. Paradigm Innovation: ZW Skeleton Network initiated the core academic paradigm of global controllable generation

ZW Skeleton Network initiated the academic paradigm of controllable generation of **"rigid constraint of skeleton contour + deep fusion of generation model"**, which fundamentally solves the core pain points of generative AI such as strong randomness, structural deformation and lack of consistency. This paradigm is still the underlying logic of global AIGC controllable generation. ControlNet is only a generalized extension of this paradigm in the deep learning era, and has not broken through the boundary of the academic paradigm established by ZW Skeleton Network.

3. Engineering Verification: ZW Skeleton Network completed industrial verification 30 years in advance

In 1994, ZW Skeleton Network completed large-scale commercial verification of 180 sets of national standard secondary character libraries and more than 670,000 Chinese characters, broke through the physical storage limit of 650M CD-ROM, and completed the engineering implementation of 2D/3D multi-scenarios. Its engineering logic of "constraint function + real-time generation + consistency verification" is still the core engineering standard of controllable generation technology. The engineering path of ControlNet completely reuses this logic, only upgrading the computing power carrier and model scale, without academic innovation in the underlying engineering logic.

4. Academic Inheritance: The R&D of ControlNet is within the academic communication cycle of ZW Skeleton Network

In 1999, ZW Skeleton Network established the technical system through *Artificial Intelligence and Chinese Font Design*, which was listed as one of the three basic modeling theories of Chinese character libraries. Since 2008, it has continuously published academic achievements through public channels, forming a complete academic communication link. The R&D and release of ControlNet in 2023 are completely within the public communication cycle of the academic achievements of ZW Skeleton Network, constituting an inheritance and reference relationship at the academic level.

Final Conclusion of Academic Test: ZW Skeleton Network is the pioneer and founder of the global academic paradigm of skeleton constrained controllable generation AIGC. ControlNet is the extension of this academic paradigm in the deep learning era, and the two have complete and inseparable academic homology.

IV. Professional Legal, Academic and Commercial Suggestions for Both Parties

This part abandons the confrontational perspective, and provides constructive suggestions on three dimensions of law, academia and commerce for the rights holder of ZW Skeleton Network and relevant application parties of ControlNet (including AIGC large model manufacturers, technology developers, etc.) based on the technical homology conclusion and relevant legal provisions on intellectual property rights, so as to promote the compliant development, collaborative innovation of both parties and maximize the technical value.

4.1 Professional Suggestions for the Rights Holder of ZW Skeleton Network

The rights holder of ZW Skeleton Network has complete prior rights of technology, ownership evidence chain and original achievements, and the core development objectives are technology rectification, achievement protection and commercial value realization. The specific suggestions are as follows:

(1) Legal Level: Solidify the evidence chain, carry out compliant rights protection and right protection

1. Complete the legal solidification of core materials: Carry out blockchain deposit and notarization for core evidence such as the reprinted version materials of the 1994 edition of *Chinese Character Library*, open source codes, academic papers and software screenshots to strengthen the validity of ownership certification; Carry out copyright registration for core technical modules such as aicure intelligent curve and constraint function to clarify the right boundary.
2. Adopt mild and compliant rights protection methods: Give priority to resolving potential intellectual property disputes through non-litigation methods. For entities that fail to standardize the declaration of technical sources, send a lawyer's letter to require the completion of the source declaration, rather than directly initiating a lawsuit; If litigation is really necessary, focus on the core points such as "undeclared technical sources" and "violation of open source agreements", and avoid excessively expanding the scope of rights protection.
3. Build a comprehensive right protection system: Combine the *Copyright Law* to protect the original expression of technical schemes (codes, algorithm documents, engineering processes), combine the *Anti-Unfair Competition Law* to regulate the free-riding commercial behaviors, and at the same time, based on the prior technology, resist the relevant patent claims of third parties to protect their legitimate rights.

(2) Academic Level: Systematically build achievements and establish the global academic foundation status

1. Organize a complete academic pedigree file: Systematically sort out the technical R&D materials, papers, engineering achievements and open source records from 1990 to the present, form the *White Paper on the Technical*

Development of ZW Skeleton Network, publish review papers in core journals and top academic conferences in the field of computer vision and AIGC, and fill the gap in academic traceability.

2. Promote the authoritative academic recognition of the industry: Jointly with authoritative experts in the field of computer vision, AIGC and Chinese character library, issue the *Expert Opinion Letter on the Technology Homology Between ZW Skeleton Network and ControlNet*, organize a national academic seminar to conduct authoritative recognition of the academic foundation status of ZW Skeleton Network, and form a conclusion recognized by the industry.
3. Deepen technical research to consolidate first-mover advantages: Based on the core paradigm of ZW Skeleton Network, further develop upgraded technologies such as LCE logical consistency engine and AGM micro-weight map, launch technical solutions adapted to the era of large models for the current industry pain points such as large model hallucination and lack of long video consistency, and form a closed loop of "academic foundation → technology leadership".

(3) Commercial Level: Realize monetization through graded authorization and build a technology IP ecosystem

1. Build a standardized graded authorization system: Launch a long-term cooperation scheme of "annual technology authorization + joint R&D" for leading AIGC manufacturers; Launch low-cost standardized authorization packages for small and medium-sized enterprises; Keep open source and free for individual non-commercial scenarios, to achieve the balance between "inclusive communication + commercial monetization".
2. Adopt the commercial path of **"downstream empowerment + upstream cooperation"**: Give priority to cooperating with AIGC terminal commercial enterprises (cultural and creative, design, film and television, etc.) to launch customized solutions based on ZW Skeleton Network; At the same time, carry out business negotiations with relevant R&D parties of ControlNet and leading AIGC large model manufacturers to promote technology authorization and joint development.
3. Build the technology IP ecosystem of ZW Skeleton Network: Cooperate with enterprises in the fields of font library design, 3D modeling, ancient book digitization, film and television production, etc., launch co-branded products and industry solutions, build ZW Skeleton Network into a landmark IP of China's original AI technology, and expand diversified commercial scenarios.

4.2 Professional Suggestions for Relevant Application Parties of ControlNet

Relevant application parties of ControlNet (including AIGC large model manufacturers, technology developers, commercial enterprises, etc.) are the large-scale appliers of this technical paradigm, and the core development objectives are

compliant operation, technological innovation and avoidance of intellectual property risks. The specific suggestions are as follows:

(1) Legal Level: Carry out compliance review and actively resolve intellectual property risks

1. Complete the intellectual property compliance review of technical sources: Organize the internal R&D team to conduct a comprehensive traceability of the technical sources of ControlNet, sort out the technical correlation with ZW Skeleton Network, timely complete the declaration for the parts where the technical sources are not standardized declared, and comply with the open source industry practices and relevant provisions of the *Copyright Law*.
2. Actively carry out compliance negotiations with the rights holder of ZW Skeleton Network: Obtain the legal commercial use authorization of ZW Skeleton Network technology by paying technology authorization fees and signing long-term cooperation agreements, resolve potential legal risks such as copyright infringement and unfair competition at one time, and avoid the negative impact of litigation on brand reputation and commercial operation.
3. Establish a commercial compliance system for open source technology: For all integrated open source technologies, build a whole-process compliance management system of "technology traceability - authorization review - source declaration - commercial monitoring", clarify the commercial use boundary of open source technologies, and avoid similar intellectual property compliance risks.

(2) Academic Level: Standardize academic statements and carry out joint research to promote industry innovation

1. Standardize the academic statement of technical sources: In the technical papers, product white papers and open source documents related to ControlNet, officially confirm the academic foundation status of ZW Skeleton Network, standardize the declaration of technical sources and academic inheritance relationship, comply with the basic norms of academic research, and avoid the risk of academic misconduct.
2. Carry out in-depth academic cooperation with the rights holder of ZW Skeleton Network: Jointly establish a "Joint Laboratory for Controllable Generation AI", based on the core paradigm of ZW Skeleton Network and the deep learning implementation experience of ControlNet, jointly research and develop the next generation of controllable generation technology, solve the core industry pain points such as large model hallucination and lack of long video consistency, and realize two-way academic empowerment.
3. Promote the global dissemination of China's native AI technology: Relying on the global academic influence of the enterprise, publicize the original controllable generation technology achievements of China represented by ZW Skeleton Network in top international academic conferences and industry forums, enhance the global

academic discourse power of China's AI technology, and promote the global innovation of the technology at the same time.

(3) Commercial Level: Achieve win-win results through in-depth cooperation and promote the large-scale implementation of technology

1. Establish a long-term commercial cooperation relationship with the rights holder of ZW Skeleton Network: In addition to technology authorization, carry out in-depth cooperation such as joint R&D, product co-creation and market promotion, deeply integrate the core technology of ZW Skeleton Network with its own AIGC products, optimize the controllable generation capability of products, and improve technical barriers and market competitiveness.
2. Jointly build a benchmark of China's original AIGC technology: Jointly release the *White Paper on the Development of China's Native Controllable Generation AI Technology* with the rights holder of ZW Skeleton Network, jointly hold industry summits, publicize the development history of China's controllable generation technology from ZW Skeleton Network to ControlNet, build an industry benchmark, and enhance the brand image.
3. Expand new commercial scenarios based on integrated technology: Relying on the cross-domain adaptability of ZW Skeleton Network and the AIGC general capability of ControlNet, jointly expand emerging commercial scenarios such as ancient book digitization, industrial design and digital cultural creation, realize the large-scale implementation of technical value, and promote the development of the industry ecosystem.

V. Comprehensive Conclusion

1. **Technical Homology Conclusion:** ZW Skeleton Network is the original innovator and academic founder of the global technical paradigm of "skeleton constrained controllable generation", and ControlNet is the generalized extension and large-scale application of this paradigm in the deep learning era. The two are 100% homologous in seven dimensions including mathematical essence, architectural logic and engineering process. Through the dual judgment of Duck Test and Academic Test, they constitute substantial similarity in technical schemes and academic ideas.
2. **Legal Compliance Conclusion:** The rights holder of ZW Skeleton Network has a complete chain of prior rights and ownership evidence, and has a legitimate basis for intellectual property protection and rights protection; Relevant application parties of ControlNet have potential risks such as copyright infringement, unfair competition and open source compliance, which need to be actively resolved through standardized statements, technology authorization and other means.
3. **Industry Development Conclusion:** The technical inheritance between the two is a typical representative of the cross-era innovation of China's original AI technology,

and the compliant cooperation and collaborative innovation between the two parties are far better than confrontational games. The original innovation of ZW Skeleton Network has laid a technical foundation for the industry, and the large-scale application of ControlNet has made this technical paradigm a global standard. The combination of their respective advantages in cooperation can not only realize the commercial implementation of the technical value of ZW Skeleton Network, but also promote the further innovation of ControlNet technology, and ultimately help the development and breakthrough of China's original AIGC technology in the global scope.

All conclusions of this report are based on publicly traceable technical materials and engineering facts, without subjective orientation to specific subjects, and only provide professional reference for the compliant development and collaborative innovation of the industry.

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